

2019

Topics

Integral calculus (2U)
Trigonometry (2U)
Polynomials (3U)
Integration by substitution (3U)

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Centre Number

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Student Number



Killara
HIGH SCHOOL

MATHEMATICS DEPARTMENT

Year 12 Mathematics Ext 1

26/2/2019

General

Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations in the examination books provided

Total Marks:40

	Question 1	/12
	Question 2	/14
	Question 3	/14
Total		/40

This question paper must not be removed from the examination room.
This assessment task constitutes 20% of the course.

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Question 1 (12 marks) Use the Question 1 Writing Booklet.

- (a) If α , β and γ are the roots of the equation $2x^3 - 3x + 1 = 0$.
What is the value of:

(i) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2

(ii) $\alpha^2 + \beta^2 + \gamma^2$ 2

- (b) Taking $x = 0.5$ radians as the first approximation, use Newton's method to find a second approximation to the root of $2x - 4\cos 3x = 0$. Give your answer to 2 decimal places. 3

- (c) Evaluate 3

$$\int_0^{\frac{\pi}{4}} \sin x \cos^3 x \, dx$$

- (d) Evaluate 2

$$\lim_{x \rightarrow 0} \frac{\tan 3x}{x}$$

End of Question 1

Question 2 (14 marks) Use the Question 2 Writing Booklet.

- (a) Evaluate $\int_0^{\frac{\pi}{3}} \sin^2 2x \, dx$ 3
- (b) Find $\int x\sqrt{4-x} \, dx$ using the substitution $u = 4 - x$ 3
- (c) Use the substitution $x = 4\tan\theta$ to find $\int \frac{dx}{(x^2 + 16)^{\frac{3}{2}}}$ 4
- (d) (i) Differentiate $x \sin x$ 2
- (ii) Hence find $\int x \cos x \, dx$ 2

End of Question 2

Question 3 (14 marks) Use the Question 3 Writing Booklet.

(a) Consider the curves $y = \sin x$ and $y = \cos 2x$ for $-\pi \leq x \leq \pi$.

(i) Show that $x = \frac{\pi}{6}$ is a root of $\sin x = \cos 2x$ 1

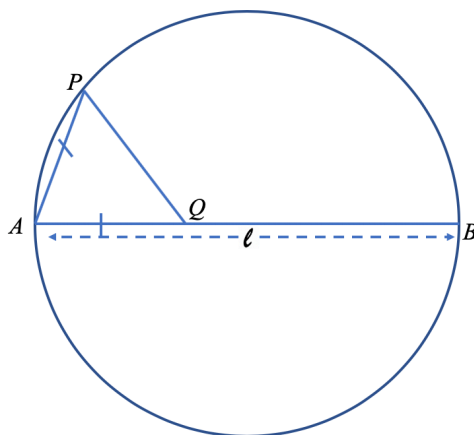
(ii) Calculate the exact area of the region bounded by the curves 3
 $y = \sin x$ and $y = \cos 2x$ for $0 \leq x \leq \frac{\pi}{2}$

(b) (i) Divide the polynomial $f(x) = 2x^4 - 10x^3 + 12x^2 + 2x - 3$ by $g(x) = x^2 - 3x + 1$. 2

(ii) Hence write $f(x) = g(x)q(x) + r(x)$, where $q(x)$ and $r(x)$ are polynomials 1
 and $r(x)$ has degree less than 2.

(iii) Hence, deduce that $f(x)$ and $g(x)$ have no zeros in common. 2

(c)



In the diagram above, P is a point on the circle with diameter $AB = \ell$. The point Q lies on the diameter such that $AP = AQ$. Let $\angle PAQ = \theta$ and let S be the area of $\triangle PQA$.

(i) Show that 2

$$S = \frac{\ell^2}{2} \cos^2 \theta \sin \theta$$

(ii) Find the maximum area of $\triangle PQA$ as P moves along the circumference of the circle. 3

END OF EXAMINATION

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End of Question 1

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- (c) Use the substitution $x = 4\tan\theta$ to find $\int \frac{dx}{(x^2 + 16)^{\frac{3}{2}}}$ 4
- (d) (i) Differentiate $x \sin x$ 2
- (ii) Hence find $\int x \cos x \, dx$ 2

End of Question 2

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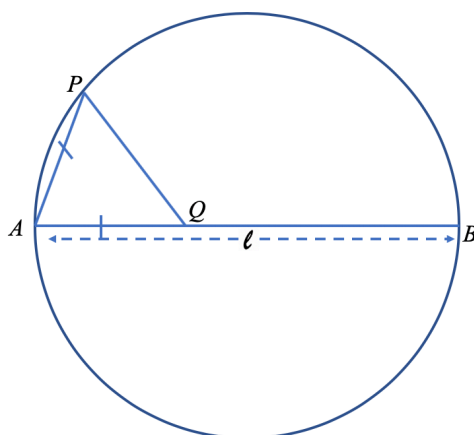
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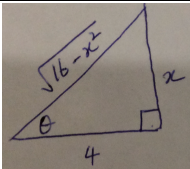
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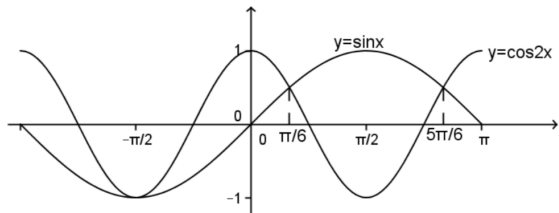
END OF EXAMINATION

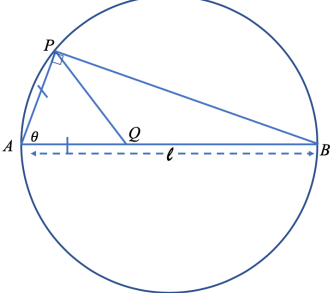
QUESTION 1

(a)(i)	$2x^3 - 3x + 1 = 0$ $\alpha + \beta + \gamma = 0. \quad (\text{Coefficient of } x^2 = 0)$ $\alpha\beta + \beta\gamma + \alpha\gamma = -\frac{3}{2}$ $\alpha\beta\gamma = -\frac{1}{2}$ $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ $= \frac{\alpha\beta + \beta\gamma + \alpha\gamma}{\alpha\beta\gamma} = \frac{-\frac{3}{2}}{-\frac{1}{2}} = 3$	1 mark for relation between roots and coefficients 1 mark: Manipulates the expression and evaluates correctly	
(a)(ii)	$\alpha^2 + \beta^2 + \gamma^2$ $= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma)$ $= 0 - 2 \times -\frac{3}{2} = 3$	2 marks: Correct answer from correct working 1 mark: correct formula, but minor error in evaluation	
(b)	$f(x) = 2x - 4 \cos 3x$ $f'(x) = 2 + 12 \sin 3x$ $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ $x_1 = 0.5 - \frac{2(0.5) - 4 \cos(3 \times 0.5)}{2 + 12 \sin(3 \times 0.5)}$ $= 0.44867 \dots$ $= 0.45 \text{ (2d.p.)}$	2 marks: correct answer from correct working 1 mark: correct formula with minor error in derivative or evaluation	
(c)	$\int_0^{\frac{\pi}{4}} \sin x \cos^3 x \, dx$ $= -\int_0^{\frac{\pi}{4}} -\sin x \cos^3 x \, dx$ $= -\left[\frac{\cos^4 x}{4}\right]_0^{\frac{\pi}{4}} = \frac{-1}{4} \left[\left(\frac{1}{\sqrt{2}}\right)^4 - 1\right]$ $= \frac{-1}{4} \left[\frac{1}{4} - 1\right] = \frac{3}{16}$	3 marks correct answer from correct working 2 marks: correct integration, minor error in evaluation 1 mark: correct integral	
(d)	$\lim_{x \rightarrow 0} \frac{\tan 3x}{x} = \lim_{x \rightarrow 0} \frac{\tan 3x}{3x} \times \frac{3x}{x}$ $= 3 \lim_{x \rightarrow 0} \frac{\tan 3x}{3x}$ $= 3$	2 marks: correct answer from correct working 1 mark: shows that $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$	

	QUESTION 2		
(a)	$\int_0^{\frac{\pi}{3}} \sin^2 2x dx = \int_0^{\frac{\pi}{3}} \frac{1 - \cos 4x}{2} dx$ $= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\frac{\pi}{3}}$ $= \frac{1}{2} \left[\frac{\pi}{3} - \frac{\sin \frac{4\pi}{3}}{4} - \left(0 - \frac{\sin 0}{4} \right) \right]$ $= \frac{\pi}{6} + \frac{\sqrt{3}}{16}$	3 marks: correct answer from correct working 2 marks: correct integral minor error in substitution 1 mark: correct expression for $\sin^2 2x$	
(b)	$\int x\sqrt{4-x} dx$ $u = 4 - x \Rightarrow x = 4 - u$ $dx = -du$ $\int x\sqrt{4-x} dx = -\int (4-u)\sqrt{u} du$ $= \int u^{\frac{3}{2}} - 4u^{\frac{1}{2}} du$ $= \frac{2}{5} u^{\frac{5}{2}} - 4 \times \frac{2}{3} u^{\frac{3}{2}} + C$ $= u^{\frac{3}{2}} \left(\frac{2}{5} u - \frac{8}{3} \right) + C$ $= (4-x)\sqrt{4-x} \left(\frac{2}{5}(4-x) - \frac{8}{3} \right) + C \quad 1 \text{ mark}$ $= -\frac{2}{15} (4-x)\sqrt{4-x} (20 - 3(4-x)) + C$ $= \frac{2}{15} (4-x)\sqrt{4-x} (8 + 3x) + C$	1 mark: correctly converts the integrand in terms of u 1 mark: correct integration 1 mark: Expresses the answer in terms of x	
(c)	$\int \frac{dx}{(x^2+16)^{\frac{3}{2}}}$ Let $x = 4\tan\theta$ Then $dx = 4 \sec^2 \theta d\theta$ 1 mark $\int \frac{dx}{(x^2+16)^{\frac{3}{2}}} = \int \frac{4 \sec^2 \theta d\theta}{(16\tan^2\theta+16)^{\frac{3}{2}}}$ $= \int \frac{4 \sec^2 \theta d\theta}{(16\sec^2\theta)^{\frac{3}{2}}} \quad 1 \text{ mark}$ $= \int \frac{4 \sec^2 \theta d\theta}{64 \sec^3 \theta}$ $= \int \frac{d\theta}{16 \sec \theta}$ $= \int \frac{1}{16} \cos \theta d\theta. \quad 1 \text{ mark}$ $= \frac{1}{16} \sin \theta + C. \quad 1 \text{ mark}$	1 mark: converts all components of the integrand in terms of θ 1 mark: expresses the integrand in the simplest form 1 mark: correct integration 1 mark: Converts the result back to x.	

	$\sin\theta = \frac{x}{\sqrt{16-x^2}}$ $\int \frac{dx}{(x^2+16)^{\frac{3}{2}}} = \frac{x}{16\sqrt{16-x^2}} + C$		
(d)(i)	$\frac{d}{dx}(x\sin x) = x\cos x + \sin x$	1 mark: for each part	
(ii)	Hence $\int x\cos x + \sin x \, dx = x\sin x + C$ $\therefore \int x\cos x \, dx - \cos x = x\sin x + C$ $\therefore \int x\cos x \, dx = x\sin x + \cos x + C$	1 mark: shows the antiderivative statement from their answer in (i) 1 mark: integrates the integrable part ($\sin x$) and hence, finds $\int x\cos x \, dx$	

	Question 3																											
(i)	$\sin \frac{\pi}{6} = \frac{1}{2}$ $\cos \left(2 \times \frac{\pi}{6} \right) = \frac{1}{2}$ $x = \frac{\pi}{6} \text{ is a root.}$	1 mark: $x = \frac{\pi}{6}$ is satisfied by both equations																										
(ii)	 Area = $\int_0^{\frac{\pi}{6}} \cos 2x - \sin x \, dx +$ $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin x - \cos 2x \, dx$ $= \left[\frac{\sin 2x}{2} + \cos x \right]_0^{\frac{\pi}{6}} +$ $\left[-\cos x - \frac{\sin 2x}{2} \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$ $= \left(\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2} - 0 - 1 \right) + \left(0 - 0 + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{4} \right)$ $= \frac{3\sqrt{3}}{2} - 1 \text{ unit}^2$	1 mark: correct graphs (rough sketch is sufficient)																										
		1 mark: correct expressions for the area formula																										
		1 mark: correct evaluation																										
(b)(i)	$\begin{array}{r} 2x^2 - 4x - 2 \\ x^2 - 3x + 1 \overline{) 2x^4 - 10x^3 + 12x^2 + 2x - 3} \\ \underline{2x^4 - 6x^3 + 2x^2} \\ -4x^3 + 10x^2 + 2x \\ \underline{-4x^3 + 12x^2 - 4x} \\ -2x^2 + 6x - 3 \\ \underline{-2x^2 + 6x - 2} \\ -1 \end{array}$ <table><tr><td></td><td>2</td><td>-10</td><td>12</td><td>2</td><td>-3</td></tr><tr><td>3</td><td></td><td>6</td><td>-12</td><td>-6</td><td>3</td></tr><tr><td>-1</td><td></td><td></td><td>-2</td><td>4</td><td>2</td></tr><tr><td></td><td>2</td><td>-4</td><td>-2</td><td>0</td><td>-1</td></tr></table>		2	-10	12	2	-3	3		6	-12	-6	3	-1			-2	4	2		2	-4	-2	0	-1	2 marks: correct division. (Students can choose to do synthetic division)		
	2	-10	12	2	-3																							
3		6	-12	-6	3																							
-1			-2	4	2																							
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		1 mark: Minor error																										

(ii)	$2x^4 - 10x^3 + 12x^2 + 2x - 3 = (x^2 - 3x + 1)(2x^2 - 4x - 2) - 1,$ $\text{Deg } r(x) < 2$	1 mark: Correct form of division transformation given									
(iii)	If $f(x)$ and $g(x)$ have a common zero, say α. Hence, $f(\alpha) = g(\alpha) = 0$ But, $f(\alpha) = g(\alpha) \cdot q(\alpha) + r(\alpha)$ $0 = 0 \times q(\alpha) + r(\alpha)$ $\Rightarrow r(\alpha) = 0$ But $r(\alpha) = 1 \neq 0$. Hence, $f(x)$ and $g(x)$ have no zeros in common.	2 marks: must show all reasoning and working to get 2 marks. (must prove conclusively using an analytic method) 1 mark: states $g(\alpha) = 0$ and attempts to prove $f(\alpha) = 0$									
(d)(i)	 $\cos \theta = \frac{AP}{l} \therefore AP = l \cos \theta$ $S = \frac{1}{2} \times l \cos \theta \times l \cos \theta \sin \theta$ $= \frac{l^2}{2} \cos^2 \theta \sin \theta$ as required. $= \frac{l^2}{2} (1 - \sin^2 \theta) \sin \theta$ $= \frac{l^2}{2} (\sin \theta - \sin^3 \theta)$	2 marks correct expression with working 1 mark: correct expression for S $S = \frac{1}{2} \times l \cos \theta \times l \cos \theta \sin \theta$									
(ii)	$\frac{ds}{d\theta} = \frac{l^2}{2} (\cos \theta - 3 \sin^2 \theta \cos \theta)$ $= \frac{l^2}{2} \cos \theta (1 - 3 \sin^2 \theta)$ $\frac{ds}{d\theta} = 0$ when $\cos \theta = 0$ or $\sin \theta = \pm \frac{1}{\sqrt{3}}$ Since $0 < \theta < 90^\circ$ $\sin \theta = \frac{1}{\sqrt{3}}$ is the only possible solution. <table border="1" data-bbox="247 1644 880 1803"> <tr> <td>$\sin \theta$</td><td>$\frac{1}{2}$</td><td>$\frac{1}{\sqrt{3}}$</td><td>$\frac{3}{4}$</td></tr> <tr> <td>$\frac{ds}{d\theta}$</td><td>+ve</td><td>0</td><td>-ve</td></tr> </table> $\therefore$ when $\sin \theta = \frac{1}{\sqrt{3}}, S$ is a maximum. $S = \frac{l^2}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{3}} \right) = \frac{l^2}{2} \left(\frac{3-1}{3\sqrt{3}} \right) = \frac{l^2}{3\sqrt{3}}$ $= \frac{\sqrt{3}l^2}{9}$ is the maximum area	$\sin \theta$	$\frac{1}{2}$	$\frac{1}{\sqrt{3}}$	$\frac{3}{4}$	$\frac{ds}{d\theta}$	+ve	0	-ve	3 marks: correct answer with correct working 2 marks: Correct solution $\sin \theta = \frac{1}{\sqrt{3}}$ And verification of max by a table. No calculation of S 1 mark correct derivative and identifying $\frac{ds}{d\theta} = 0$ for $\cos \theta = 0$ or $\sin \theta = \pm \frac{1}{\sqrt{3}}$	
$\sin \theta$	$\frac{1}{2}$	$\frac{1}{\sqrt{3}}$	$\frac{3}{4}$								
$\frac{ds}{d\theta}$	+ve	0	-ve								